

HINTS AND SOLUTIONS

- Q.1 C
Q.2 A
Q.3 C
Q.4 A
Q.5 B
Q.6 B
Q.7 D
Q.8 B
Q.9 B
Q.10 B

Sol. Initially, charge $q = CV$

and energy stored in capacitor $= \frac{1}{2} CV^2$

Finally, charge $q_2 = C \times 2V = 2CV$

Charge flown through the battery is

$$q' = q_2 - (-q) = 3CV$$

$\therefore$ Energy drawn from battery

$$E_b = q' \times V' = 3CV \times 2V = 6CV^2$$

$$\text{Energy stored in capacitor} = \frac{1}{2} C(2V)^2$$

$$E_2 = 2CV^2$$

$$\text{Change in potential energy of capacitor} = 2CV^2 - \frac{1}{2} CV^2$$

$$\Delta E_c = \frac{3}{2} CV^2$$

$$\text{Energy lost in heat} = E_H = 6CV^2 - \frac{3}{2} CV^2 = \frac{9}{2} CV^2$$

$$\text{The ratio of } \frac{E_H}{E_2} = \frac{9}{4} = 2.25$$

- Q.11 A

Sol. $E = \frac{V}{d} \quad C = \frac{\epsilon_0 A}{d}$

$$V = Ed \Rightarrow d \text{ is more} \Rightarrow V \text{ is more}$$

$$U = \frac{1}{2} CV^2 \Rightarrow V \text{ is more \& } U \text{ is more}$$

Q.12 D

$$\begin{aligned}\text{Sol. } V &= -\int_0^d \frac{aQ}{\epsilon_0} \left(1 + \frac{x^2}{d^2}\right) dx \\ &= -\frac{aQ}{\epsilon_0} \left[d + \frac{d}{3}\right] = \frac{4aQd}{3\epsilon_0}\end{aligned}$$

Q.13 C

$$\text{Sol. } C = \frac{Q}{V} = \frac{3\epsilon_0}{4ad}$$

Q.14 C

$$\text{Sol. } \frac{\frac{1}{2}\epsilon_0 E_0^2}{\frac{1}{2}\epsilon_0 E_p^2} = \frac{E_0^2}{E_p^2} = \frac{\left(\frac{aQ}{\epsilon_0}\right)^2}{\left(\frac{aQ}{\epsilon_0}\right)^2} = (1+1)^2 = \frac{1}{4}$$

Q.15 ABD

Q.16 BCD

Q.17 AD

$$\text{Q.18 [Ans. } C_{AD} = \frac{48}{7}\pi\epsilon_0 R]$$

$$\text{Q.19 [Ans. } \frac{11C}{15}]$$

$$\text{Q.20 [Ans. } \frac{12}{11}\mu\text{C}, 3.34\mu\text{J}]$$

$$\text{Q.21 Ans (a) } 20\mu\text{C}, \text{ (b) } 0.3\text{ mJ}, \text{ (c) } 0.6\text{ mJ}, \text{ (d) } 60\mu\text{C}$$

